Hany Hamed Aly

Interested in Robot Learning, Multi-Robot Systems & Sim2Real

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Education

Innopolis University Innopolis, Russia

*Bachelor of Computer Science (Robotics track) with **Full scholarship*** *Aug 2018 – June 2022*

- Outstanding Contribution to Science at Innopolis University during 2020/2021 (Certificate)
- Outstanding Achievements at Innopolis University for Fall 2020 (Diploma)

CGPA: 4.65/5 (Original Transcript)

Research Experience

Bachelor Thesis | *In progress* May 2021 – June 2022

- Topic: **Learning behavioural strategies for a multi-robot system in a predator-prey environment using Reinforcement Learning**
- Keywords: reinforcement learning, predator-prey, self-play, multi-agent
- Progress: link
- Supervised by: Prof. Stefano Nolfi & Prof. Alexandr Klimchik

Laboratory Assistant June 2021 – Present

Unmanned Technology Laboratory, Innopolis University *Innopolis, Russia*

- Integrating and developing ROS packages of different modules for drones (Gimbal control, RTK GPS, ...etc)
- Integrated livox lidar with a drone for outdoor mapping
- Developed a handheld lidar for indoor/outdoor mapping for an industrial partner (results)

Undergraduate Research Assistant Aug 2019 – April 2021

Center for Technologies in Robotics and Mechatronics Components, Innopolis University *Innopolis, Russia*

- Implemented a gym environment for a Tensegrity hopper
- Developed experiments using ARS RL algorithm to learn a stabilizing control policy for a tensegrity hopper
- Designed and implemented a contactless differentiable physics simulator for tensegrity robots using Taichi
- Performed research on sim2real transfer for a three-prism tensegrity robot

Publications & in progress manuscripts

- G. Kulathunga, H. Hamed, D. Devitt and A. Klimchik, "Optimization-based Trajectory Tracking Approach for Multi-rotor Aerial Vehicles in Unknown Environments" [Accepted in IEEE Robotics and Automation Letters (RA-L) 2022]
- V. Kurenkov, H. Hamed, and S. Savin, "Learning stabilizing control policies for a tensegrity hopper with augmented random search," in *2020 International Conference on Industrial Engineering, Applications and Manufacturing (ICIEAM)*, pp. 1–5, IEEE, 2020 (code, paper)
- L. Vorochaeva, S. Savin, H. Hamed, and A. M. Leon, "Analysis of algorithms for controlling the length of crawling robot modules," in *2020 4th Scientific School on Dynamics of Complex Networks and their Application in Intellectual Robotics (DCNAIR)*, pp. 257–260, IEEE, 2020 (paper)
- L. Vorochaeva, S. Savin, and H. Hamed, "Lateral gait analysis of a crawling robot by means of controlling the lengths of links and friction in the supports," in *2020 International Conference Nonlinearity, Information and Robotics (NIR)*, pp. 1–6, IEEE, 2020 (paper)
- H. Hamed, V. Kurenkov and S. Savin, "Differentiable Tensegrity Simulator with Sim2Real Experiments" [Manuscripts in preparation]
- H. Hamed, S. Savin and L. Vorochaeva "A Brief Survey on Locomotion Control Methods For Crawling Robots" [Manuscripts to be submitted in Nonlinearity, Information and Robotics (NIR) 2022]

Selected Awards & Certificates & Summer Schools

Online Asian Machine Learning School (OAMLS 2021, certificate) November 2021

- This school covered multiple topics in Machine learning: computer vision, deep learning, NLP, representation learning, transfer learning, reinforcement learning, causality and meta learning.

European Agent Systems Summer School (EASSS 2021) July 2021

- The summer school explored different topics related to multi-agent systems around cooperative systems, reinforcement learning, game theory, modeling and simulation.

3rd place DOTS competition from Bristol Robotics Lab and Toshiba (Certificate) June 2021

- The competition was about developing an algorithm for multi robots to collect food items and return it to the storage in a gazebo simulated environment.

Winter School on Machine Learning in Robotics (Certificate of Participation) Dec 2020

- This school explored the intersection of machine learning in robotics in areas of reinforcement learning, and computer vision

Skills & Tools

- ROS, Python, C/C++, PyTorch, Docker, git, Linux

Working Experience

Technical Intern July 2019 – Aug 2019 - [Doc]
Copter Express *Moscow, Russia*

- Had a training about main components and construction of drones
- Had a training on programming COEX's drones (ROS, px4 and indoor navigation using Aruco markers)
- Developed a human pose estimation with a webcam using Tensorflow.js to control a drone
- Integrated RTAP-Map SLAM algorithm with COEX's drone using Intel Real-Sense T265 and D435 cameras

Projects

Drone Detection (Computer Vision project course) | Link | *PyTorch* September 2021

- Collected and labeled a diverse dataset
- Implemented training scripts for Faster R-CNN pretrained models from dectron2

Human Pose Estimation Drone Control | Link | *Python, TF.js, ROS* July 2019

- Explored and tested different human pose estimation modules
- Integrated a drone controller with human pose estimation module built with TensorFlow.js

Extracurricular activities

Judge at WRO International Finals [Advanced Robotics Challenge - ARC] | Certificate 2019

- Developed the game rules for the international competition
- Judged the Russian qualifications as the Head judge and the International competition as a judge
- Collaborated in the development of a playground generator for the competition using C++ and Javascript

Participant in RoboCup Junior Egypt Open Weight Soccer Category (2nd place) 2018

- Developed ball detection using OpenCV and the robot's game controller algorithm
- Developed PCB boards for the robot